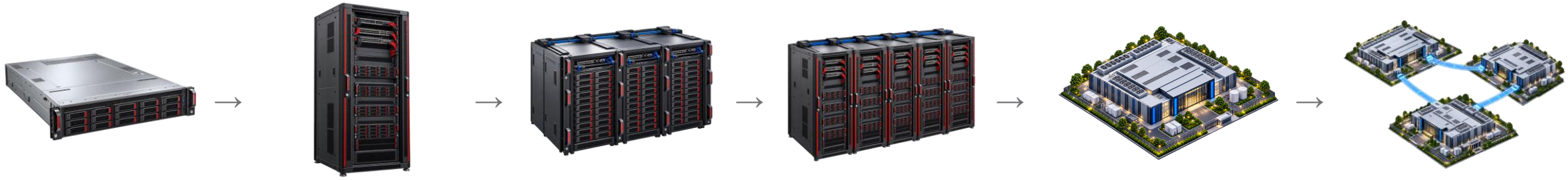
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Evolving the Infrastructure Stack
Presented by AMI

Rack Scale Designs, Open Firmware,
and Intelligent Sustainability

The Unit of Compute Keeps Expanding

The operational boundary is moving from individual platforms to connected infrastructure.



SERVER → RACK → POD → MULTI-POD → DATA CENTER → MULTI-DC

Increasing scope → more components, more failure domains, more coordination

FABRIC EVOLUTION	SCALE UP	SCALE OUT	SCALE ACROSS
	tightly coupled domains	across racks / clusters	across sites / domains

The fabric is already laying the groundwork for increasingly distributed compute domains.

As compute domains expand, the surrounding infrastructure has to behave as a coordinated system.

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Firmware Orchestration Must Scale With the System

Firmware stays close to hardware. The complexity of coordinating it expands with the operational domain.

ORCHESTRATION SCOPE



SERVER → 

RACK → 

POD → 

DATA CENTER → 

MULTI-DC

Policy • updates • validation • security posture • recovery • lifecycle

Firmware stays close to hardware. Coordination expands with the operational domain.

FIRMWARE DOMAINS



UEFI / BOOT

EDK2 + platform firmware



BMC / MGMT

OpenBMC + Redfish



SECURITY

RoT / policy / attestation



DEVICE FW

NIC • GPU • FPGA • storage

As scale increases, standardization and firmware expertise become part of the control-plane problem.

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Open Is How Heterogeneous Infrastructure Scales

Operators need a common foundation across vendors, device types, and host silicon architectures—before adding their own vertical integration.

OPERATOR INTEGRATION | **POLICY** | **AUTOMATION** | **ORCHESTRATION** | **OPTIMIZATION**

OPEN FOUNDATION + COMMON ABSTRACTIONS



UEFI / EDK2



OpenBMC



Redfish / common APIs



OpenRMC / rack APIs

Common contracts for discovery • telemetry • control • lifecycle

DIVERSE INFRASTRUCTURE



ACCELERATORS



NIC + STORAGE



POWER + COOLING



HOST Si + ARCH



VENDORS + GENERATIONS

Normalize below. Differentiate above.

WHY OCP? | **Build the common contract in OCP — define what must be common, leave room to differentiate.**

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Open Foundations Still Need Production Engineering

Open accelerates innovation; production infrastructure still requires disciplined integration.



SILICON + DEVICE ENABLEMENT



SECURITY + TRUST



PLATFORM INTEGRATION



VALIDATION + QUALIFICATION

OPEN FOUNDATIONS

Open source + common abstractions



INTEROPERABILITY



TELEMETRY + OBSERVABILITY



LIFECYCLE + UPDATES



SUPPORTABILITY AT SCALE

Open gets you started. Engineering makes it deployable, secure, interoperable, and supportable.

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The Future Control Plane Is Hierarchical — and Secure by Design

Use a common management model at every layer—and treat every management boundary as a trust boundary.

OCP WORKSTREAM • SECURE CONTROL & MANAGEMENT PLANE

GLOBAL POLICY • INTENT • AUTHORIZATION • COORDINATION



SERVER → RACK → POD → ROW → ROOM → DATA CENTER → MULTI-DC

SECURE CONTROL PLANE • identity • authorization • trusted state • secure update • recovery

← TRUSTED STATE • FILTERED TELEMETRY • ATTESTATION • INSIGHTS

ACT LOCALLY Keep time-critical decisions and immediate recovery close to hardware.

COORDINATE GLOBALLY Aggregate, authorize, downsample, and optimize across broader domains.

FPGA-based secure control can provide hardware-rooted trust, deterministic control, and parallel telemetry.

Make decisions at the right layer. Enforce trust across every layer.

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Telemetry Is the Beginning, Not the Destination

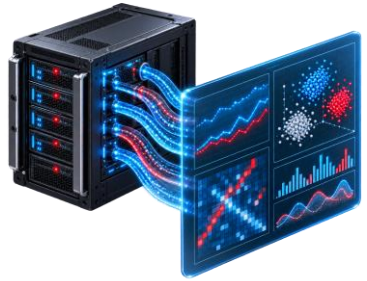
Telemetry becomes valuable when analysis turns it into prediction, optimization, and action.



OBSERVE

Power • thermal • flow
firmware • device health

RAW + LOCAL TELEMETRY



ANALYZE + UNDERSTAND

Correlate state + context
trends • policy • history

CONTEXT AT THE RIGHT LAYER

ENABLES



PREDICT Failures • thermal pressure • capacity

FORECAST WHAT COMES NEXT



OPTIMIZE Placement • power • cooling • maintenance

IMPROVE HOW RESOURCES ARE USED



ACT Boot • update • recover • isolate

CHANGE THE SYSTEM NOW

Action changes state → new telemetry

Examples: predictive maintenance • power + cooling optimization • workload placement • Rack Boot / recovery • firmware fleet coordination

The goal is not more telemetry. It is better decisions at the right layer.

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One Architecture, Many Layers of Management

AMI technologies illustrate how open foundations and secure hierarchical management can span firmware through data-center operations.

AMI EXPERTISE

	**MULTI-DATA CENTER**	AMI DCM — cross-site orchestration + global optimization
	**DATA CENTER / ROOM / ROW**	AMI DCM — hierarchical policy + filtered telemetry
	**RACK / POD**	AMI RM — OpenRMC-compliant rack / pod management + orchestration
	**FW RACK MANAGEMENT**	OneTree / OpenBMC-based firmware rack management
	**PLATFORM MANAGEMENT**	OneTree / OpenBMC — BMC manageability + telemetry + lifecycle
	**PLATFORM TRUST / PRoT**	Tektagon firmware + Lattice FPGA-based platform root of trust
	**BOOT + PLATFORM FW**	UEFI / EDK2-based open firmware + platform enablement

Open firmware • platform trust • hierarchical orchestration

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Call to Action: Bring Firmware Into the Next-Generation Data Center

The architecture is changing. Firmware can be part of how you design, secure, and operate it.

COME BY THE AMI & LATTICE BOOTHS (A54 & A53)

Discuss how firmware can be part of your next-generation data center.

- **OPEN FIRMWARE** — UEFI / EDK2 • OpenBMC • OpenRMC
- **SECURE CONTROL** — Platform Trust • Secure Management • Telemetry
- **HIERARCHICAL ORCHESTRATION** — Server → Rack → Pod → DC → Multi-DC

Bring your next architecture challenge.



From platform firmware to the data center control plane.

COME TALK WITH AMI — AND HELP BUILD THE COMMON CONTROL PLANE IN OCP.

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OPEN. COORDINATED. SECURE. INTELLIGENT.



Come build the future of the data center control plane with us — in the OCP community.

AMI • OCP Global Summit 2026